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22/07/2025

DIVING SERVICE REPORT UNIT 2A STOP LOG INSPECTION.



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INTRODUCTION.

On **22nd July 22, 2025**, a diving operation was carried out at the top dam unit 2A of **Bujagali Hydropower Plant**, to investigate and assess leakage through the stop logs recently installed. The inspection revealed that the stop logs did not seal completely due to foreign materials, specifically, pieces of logs/ branches and a small stone lodged underneath them. These obstructions created an approximately 2- 3 cm wide gap, which allowed water to pass through. A continuous seepage of water was observed. Stop logs are intended to form a watertight barrier, but leakage can compromise operational safety and efficiency. Bujagali HPP engaged the diving team to conduct an underwater inspection to identify the cause of the leakage.

SCOPE.

- i. To conduct a visual and physical inspection of the stop log seating area.
- ii. To determine and document the cause of water leakage.
- iii. To assess any other conditions affecting the performance of the stop logs.
- iv. To provide professional recommendations for remedial action.

METHODOLOGY.

The following steps were taken:

- Lock-out and isolation procedures were confirmed before commencing.
- Equipment checks and safety briefings were conducted prior to entry.
- The diver visually inspected the contact surface between the stop logs and the sill/ bottom plate. Manual probing was performed to feel for suction or any irregularities.
- A dive supervisor and Bujagali team monitored all activities from the surface.

PHOTOGRAPHIC SECTION.

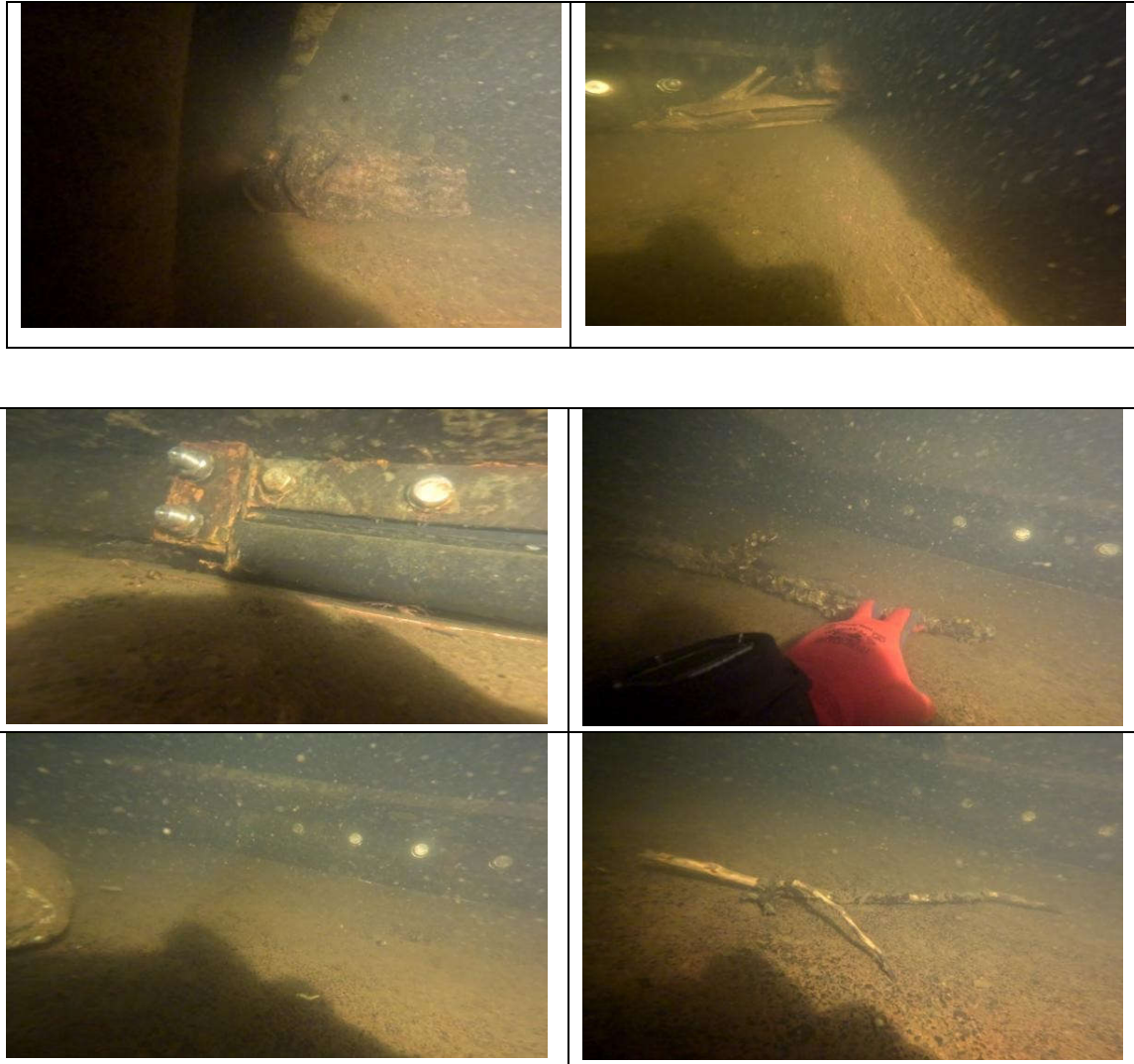


Table 1. Showing some of the foreign materials that lodged underneath the stop log.

OBSERVATIONS.

- Pieces of wooden logs and one small stone were found beneath the installed stop logs.
- These obstructions created an uneven seating surface, preventing the stop logs from sitting flush on the sill.
- An approximately 2-3 cm wide gap was measured along portions of the stop log base, which allowed continuous water flow.
- The stop logs themselves were in good physical condition with no structural damage observed.
- There were few accumulation of logs and stones in the immediate area which were pushed away by the diver.

CHALLENGES.

- Underwater visibility was reduced due to suspended particles.
- Water Flow: A mild but noticeable current required divers to maintain firm positioning during inspection.

RECOMMENDATION.

1. Carefully remove all pieces of wood and the stone lodged beneath the stop logs to ensure a clear and smooth seating surface.
2. After cleaning, re-install or re-seat the stop logs to achieve a complete seal.
3. Conduct another dive inspection after re-seating to confirm the effectiveness of the seal.

CONCLUSION.

The diving operation successfully identified the possible root cause of the leakage. The stop logs are not defective; however, foreign materials (pieces of logs and a small stone) beneath them created an approximate 2 cm gap, resulting in water leakage. Prompt removal of the debris and proper re-seating of the stop logs will restore a watertight seal and ensure reliable operation.